1 The Model

1.1 Demographics

We adopt a life cycle model with intergenerational altruism. Households go through two stages of life, young and old age. A young person faces a constant probability of aging during each period $(1 - \pi_y)$, and an old person faces a constant probability of dying during each period $(1 - \pi_o)$. When an old person dies, his offspring enters the model, carrying the assets bequeathed to him by the parent. Appropriately parameterized, this framework generates households for which the average lengths of the working period and the retirement period are realistic. Our model period is one year.

There is a continuum of households of measure one. The households are subject to idiosyncratic shocks, but there is no aggregate uncertainty, as in **?**.

1.2 Preferences

The household's utility from consumption is given by

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}. \tag{1}$$

The households discount the future at rate β, and, in addition, they discount the utility of their offspring at rate η. In the purely life cycle version ($\eta = 0$), individuals put no weight on the utility of their descendants. In the perfectly altruistic version ($\eta = 1$), individuals care about their descendants as much as themselves. We assume exogenous labor supply.

1.3 Technology

Each person possesses two types of ability: entrepreneurial ability (θ), the capacity to invest capital more or less productively, and working ability (y), the capacity to produce income out of labor.

Entrepreneurs can borrow and invest capital in a technology whose return depends on their own entrepreneurial ability. When the entrepreneur invests k, the production is given by θk^ν, where $\nu \in [0, 1]$. Entrepreneurs thus face decreasing returns from investment, since their managerial skills become gradually stretched over larger projects (as in **?**).

Workers can save (but not borrow) at a riskless, constant rate of return. The nonentrepreneurial sector is represented by a standard Cobb-Douglas production function:

$$F(K_c, L_c) = AK_c^\alpha L_c^{1-\alpha}, \tag{2}$$

where K_c and L_c are the total capital and labor inputs in the nonentrepreneurial sector and A is a constant. In both sectors, capital depreciates at a rate δ.

1.4 Credit Market Constraints

As in **?** and **?**, the borrowing constraints are endogenously determined in equilibrium and stem from the assumption that contracts are imperfectly enforceable.

We assume that the entrepreneurs who borrow can either invest the money and repay their debt at the end of the period or run away without investing it and be workers for one period. In the latter case, they retain a fraction f of their working capital k (which includes their own assets and borrowed money), and their creditors seize the rest. Hence, the entrepreneur's assets act as collateral, although the loan need not be fully collateralized.

1.5 Households

Each young individual starts the period with assets a, entrepreneurial ability θ, and worker ability y and chooses whether to be an entrepreneur or a worker. The young's value function is

$$V(a, y, \theta) = \max\left\{V_e(a, y, \theta),\ V_w(a, y, \theta)\right\}, \tag{3}$$

where $V_e(a, y, \theta)$ is the value function of a young entrepreneur:

$$V_e(a, y, \theta) = \max_{c,k,a'}\left\{u(c) + \beta\pi_y EV(a', y', \theta') + \beta(1-\pi_y)EW(a', \theta')\right\}, \tag{4}$$

$$a' = (1-\delta)k + \theta k^{\nu} - (1+r)(k-a) - c, \tag{5}$$

$$u(c) + \beta\pi_y EV(a', y', \theta') + \beta(1-\pi_y)EW(a', \theta') \geq V_w(f \cdot k, y, \theta), \tag{6}$$

$$a \geq 0. \tag{7}$$

The enforcement constraint (**??**) ensures that the entrepreneur prefers repaying to defaulting: in case of default, the entrepreneur retains fraction f of working capital and becomes a worker.

The value function for the young worker is

$$V_w(a, y, \theta) = \max_{c,a'}\left\{u(c) + \beta\pi_y EV(a', y', \theta') + \beta(1-\pi_y)W_r(a')\right\} \tag{8}$$

subject to $a' = (1+r)a + (1-\tau)wy - c$ and $a \geq 0$, where w is the wage and τ is a proportional payroll tax financing old-age social security.

The old entrepreneur can choose to continue the entrepreneurial activity or to retire:

$$W(a, \theta) = \max\left\{W_e(a, \theta),\ W_r(a)\right\}. \tag{9}$$

1.6 Equilibrium

Let $\mathbf{x} = (a, y, \theta, s)$ be the state vector for an individual, where s distinguishes young workers, young entrepreneurs, old entrepreneurs, and old retired. A stationary equilibrium is given by a risk-free interest rate r, wage rate w, and tax rate τ; allocations $c(\mathbf{x})$,

$a(\mathbf{x})$, occupational choices, and investments $k(\mathbf{x})$; and a constant distribution $m^*(\mathbf{x})$, such that:

- The functions c, a, and k solve the maximization problems described above.
- The capital and labor markets clear.
- The wage and interest rates are given by the marginal products of each factor of production.
- The social security budget constraint is balanced period by period.
- The distribution m^* is the invariant distribution for the economy.